

Four equal dice with every sufficiently large admissible face count

Research certificate, 30 September 2026

This note gives a small explicit certificate. No claim of literature priority is made. Write a die collection as a word, assigning increasing integer labels to successive positions.

Proposition 1. *Four permutation-fair equal-sided dice exist with M faces each for every multiple M of 6 with $M \geq 12$.*

Proof. Define three palindrome blocks

$$A = 12033021, \quad B = 32011023, \quad C = 02133120.$$

The words

$$W_{12} = ABCCBA, \quad W_{18} = ABCCABBCA$$

have respectively 12 and 18 occurrences of each of the four letters. Their distinct-letter subsequence counts are as follows; each entry in a column applies to every ordered pattern of that length.

	1	2	3	4
W_{12}	12	72	288	864
W_{18}	18	162	972	4374

These integers can be checked directly by the following recurrence. For each pattern $p_1 \cdots p_k$, initialize $d_0 = 1$ and $d_j = 0$ for $1 \leq j \leq k$. Read the word from left to right. At a letter x , update $d_j \leftarrow d_j + d_{j-1}$ whenever $p_j = x$, processing indices in decreasing order. The final d_k counts the desired subsequences. The companion program and independent certificate check all 64 nonempty patterns. The displayed fourth-order counts are respectively $12^4/24$ and $18^4/24$, proving fairness.

Concatenating equal-composition fair words preserves fairness: for a fixed length- k pattern, selecting its first j letters from a block with count a and its remaining letters from one with count b gives $a^j b^{k-j} / (j!(k-j)!)$ occurrences. Summation over j gives $(a+b)^k / k!$. Every integer $M/6 \geq 2$ is $2u+3v$ with nonnegative integers u, v . Concatenating u copies of W_{12} and v copies of W_{18} proves the assertion. $\square$

In particular, the tempting stronger condition $v_p(M) \geq \lfloor n/p \rfloor$ for equal-sided fair dice is false: $n = 4$, $M = 18$, and $p = 2$ give a counterexample. The example makes no assertion about the exceptional count $M = 6$.

Reproducibility. `search_palindrome_18.cpp` enumerates all triples of relabelled palindrome blocks, retains the 192 three-wise fair triples, and finds three of their 24 distinct fourth-order defect vectors summing to zero. `four_dice_18_faces_root_check.json` records a separate Python integer dynamic-programming verification of the resulting word.